

$$D = D_1 \cup D_2$$

$D_1 \rightarrow$ predictive model $\hat{f}_{n_1}$

$D_2 \rightarrow$ cal data for $i = 1 \dots n_2$

||
| D_2 |

$$R_i = |y_i - f(x_i)|$$

$\hat{I}_n = \lceil (1-\alpha)(n_2+1) \rceil$ smallest of $R_1 - R_{n_2}$

$$\hat{C}(X_{n+1}) = (\hat{f}_{n_1}(X_{n+1}) - \hat{I}_n, \hat{f}_{n_1}(X_{n+1}) + \hat{I}_n)$$

Assumptions?

- Model-free for $\hat{f}_{n,1}(\cdot)$
- Strong case. $\underbrace{D_1, D_2}_{\{(X_i, Y_i)\}_{i=1}^n} (X_{n+1}, Y_{n+1}) \stackrel{iid}{\sim} P_{X \times Y}$
- weak case $\underbrace{D_1, D_2}_{\{(X_i, Y_i)\}_{i=1}^n} (X_{n+1}, Y_{n+1})$
"exchangeable"

$$D_1, D_2 \sim P_{X \times Y}^{(1)} \quad (X_{n+1}, Y_{n+1}) \sim P_{X \times Y}^{(2)}$$

Covariate shift:

$$P_{X \times Y}^{(1)} = P(Y|X) P^{(1)}(X)$$

$$P_{X \times Y}^{(2)} = P(Y|X) P^{(2)}(X)$$

Exchangeability and Permutations

Definition

Definition 2.1: Exchangeability

Let $Z_1, \dots, Z_n \in \mathcal{Z}$ be random variables with a joint distribution. We say that the random vector $(Z_1, \dots, Z_n)$ is *exchangeable* if, for every permutation $\sigma \in \mathcal{S}_n$,

$$(Z_1, \dots, Z_n) \stackrel{d}{=} (Z_{\sigma(1)}, \dots, Z_{\sigma(n)}),$$

where $\stackrel{d}{=}$ denotes equality in distribution, and $\mathcal{S}_n$ is the set of all permutations on $[n] := \{1, \dots, n\}$.

iid is a special case.

- The elements of an exchangeable sequence are identically distributed, but not necessarily independent.
- $Z_1, \dots, Z_n$ are sampled uniformly without replacement from a finite set $\{z_1, \dots, z_N\} \subseteq \mathcal{Z}$.
- $Z_1, \dots, Z_n$ are drawn i.i.d. from a distribution P on $\mathcal{Z}$.

Example

(z_1, z_2)

$z_i \in \{0, 1\}$

- This joint distribution is exchangeable
- Cannot be expressed either via sampling without replacement or via i.i.d. sampling.

$$\frac{1}{8}\delta_{(0,0)} + \frac{3}{8}\delta_{(0,1)} + \frac{3}{8}\delta_{(1,0)} + \frac{1}{8}\delta_{(1,1)}.$$

$$P(z_1, z_2) = P(z_2, z_1)$$

$\Leftrightarrow$

$$p(0, 0) = p(1, 1) = \frac{1}{8}, \quad p(0, 1) = p(1, 0) = \frac{3}{8}.$$

$\Rightarrow$ PMF

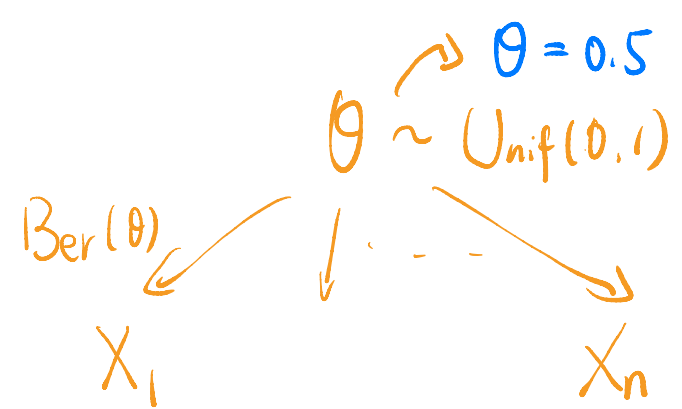
Example

$$P(z_1=1 | z_2=1) \neq P(z_1=1)$$

Let $\theta \sim \text{Uniform}(0, 1)$, and assume $(X_i | \theta) \stackrel{\text{i.i.d.}}{\sim} \text{Bernoulli}(\theta)$, $i = 1, 2, \dots, n$. Then random variables $X_1, X_2, \dots, X_n$ are exchangeable.

From the definition of the conditional distribution, given θ , the joint distribution of $X_1, X_2, \dots, X_n$ is shown as:

$$\mathbb{P}(X_1 = x_1, \dots, X_n = x_n | \theta) = \prod_{i=1}^n \theta^{x_i} (1 - \theta)^{1-x_i}.$$



$$X_1 \dots X_n \mid \theta = 0.5$$

Linear Regression

$$Y_i = X_i \beta + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma^2)$$

$$X_i, Y_i \stackrel{\text{iid}}{\sim} P_{X, Y}$$

Bayesian linear regression

$$Y_i = X_i \beta + \varepsilon_i$$

$$\varepsilon_i \sim N(0, \sigma^2)$$

$$\sigma^2 \sim \text{Gamma}(\alpha, \beta)$$

$\delta_x(A)$ is delta measure

$$\delta_x(A) = \mathbb{1}(x \in A) = \begin{cases} 1 & x \in A \\ 0 & x \notin A \end{cases}$$

$$\delta_{(0,0)}(x) = \begin{cases} 1 & x = (0,0) \\ 0 & x \neq (0,0) \end{cases}$$

Substituting this into the integral, we get:

$$\begin{aligned}
 \mathbb{P}(X_1 = x_1, \dots, X_n = x_n) &= \int_0^1 \mathbb{P}(X_1 = x_1, \dots, X_n = x_n, \theta) d\theta \\
 &= \int_0^1 \mathbb{P}(X_1 = x_1, \dots, X_n = x_n \mid \theta) p(\theta) d\theta \\
 &= \int_0^1 \prod_{i=1}^n \theta^{x_i} (1 - \theta)^{1-x_i} p(\theta) d\theta \\
 &= \int_0^1 \theta^{\sum_{i=1}^n x_i} (1 - \theta)^{n - \sum_{i=1}^n x_i} d\theta.
 \end{aligned}$$

The random variables $X_1, X_2, \dots, X_n$ are exchangeable but not i.i.d. Their joint distribution depends only on the sum of X_i rather than their specific order.

Alternative characterizations of exchangeability

- Exchangeability can be formally described in a number of different ways.

Symmetry of the joint density.

Discrete case

✓ PMF

$$p(z_1, \dots, z_n) = p(z_{\sigma(1)}, \dots, z_{\sigma(n)}) \text{ for all } z_1, \dots, z_n \in \mathcal{Z} \text{ and for all } \sigma \in S_n.$$

Continuous case

↪ pdf

$$f(z_1, \dots, z_n) = f(z_{\sigma(1)}, \dots, z_{\sigma(n)}) \text{ for almost every } (z_1, \dots, z_n) \in \mathbb{R}^n \text{ and for all } \sigma \in S_n.$$

rank $z_1, \dots, z_n$ small - large.

Conditioning on the order statistics.

$$z_{(1)} \leq z_{(2)} \leq \dots \leq z_{(n)}$$

- Conditioning on the order statistics, the random vector $(Z_1, \dots, Z_n)$ is equally likely to be any one of the $n!$ unique permutations of the order statistics.

$$(Z_1, \dots, Z_n) \mid (Z_{(1)}, \dots, Z_{(n)}) \sim \frac{1}{n!} \sum_{\sigma \in S_n} \delta_{(Z_{(\sigma(1))}, \dots, Z_{(\sigma(n))})}$$

Conditional on the unordered collection of values in the sequence, the order in which they appear is simply a random shuffle.

- Under exchangeability of $Z_1, \dots, Z_n$, it must hold that

"uniform distribution" with

$n!$

prob $\frac{1}{n!}$ for each $(z_{\sigma(1)}, z_{\sigma(2)}, \dots, z_{\sigma(n)})$

PMF

$$Z_i \mid (Z_{(1)}, \dots, Z_{(n)}) \sim \frac{1}{n} \sum_{j=1}^n \delta_{Z_{(j)}}$$

$Z_i \mid 1, 3, 4, 10$

Z_i

z	$P(z)$
1	$1/n$
3	$1/n$
4	$1/n$
$\vdots$	$\vdots$
10	$1/n$

(2.2)

(i.e., each individual entry Z_i is equally likely to be any one of the n order statistics),

$$\mathbb{P}(Z_i \leq Z_{(k)}) \geq k/n,$$

Moreover, if the values $Z_1, \dots, Z_n$ are distinct almost surely, then this becomes an equality.

$$P(Z_i = Z_{(k)}) = \dots = P(Z_i = Z_{(n)}) = \frac{1}{n}$$

Conditioning on the empirical distribution.

In the real-valued setting discussed above, we saw that each entry Z_i can be viewed as a random draw from the values $Z_{(1)}, \dots, Z_{(n)}$. In fact, this intuition can be extended to the general setting, beyond the case $Z = \mathbb{R}$.

Define

$$\hat{P}_n = \frac{1}{n} \sum_{i=1}^n \delta_{Z_i},$$

$$Z_{(1)} \text{ --- } Z_{(n)}$$

$\uparrow$
 $Z_{(k)}$

which is the empirical distribution of the random vector $(Z_1, \dots, Z_n)$.

Proposition 2.2: Exchangeability and the empirical distribution

Let $(Z_1, \dots, Z_n) \in \mathcal{Z}^n$ be an exchangeable random vector, and let $\hat{P}_n$ be the empirical distribution of this vector. Then for all $i \in [n]$,

$$Z_i \mid \hat{P}_n \sim \hat{P}_n,$$

i.e., if we condition on $\hat{P}_n$, then $\hat{P}_n$ is itself the conditional distribution of Z_i .

The proof of Proposition 2.2 relies on the fact that $Z_1, \dots, Z_n$ are exchangeable even after conditioning on $\hat{P}_n$. In fact, this is a special case of the following lemma, which verifies that exchangeability continues to hold after conditioning on any symmetric function of $Z_1, \dots, Z_n$.

Empirical distribution

$$\underline{Z \sim P}$$

z	1	2	3
$P(Z=z)$	$1/2$	$1/2$	$1/4$

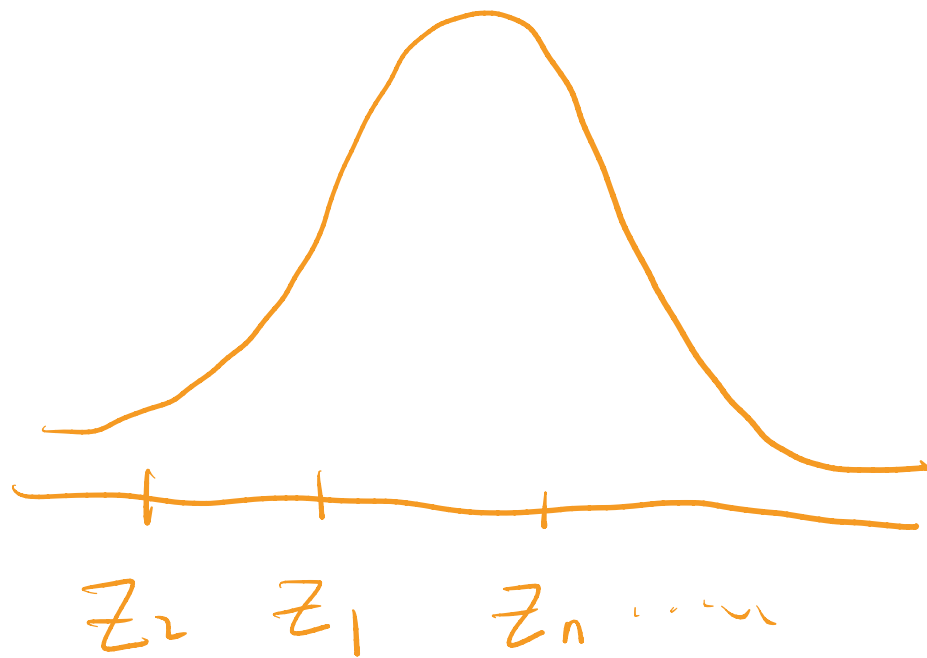
$$Z_1, \dots, Z_n \stackrel{\text{iid}}{\sim} P$$

$$\hat{P}_n = \frac{1}{n} \sum_{i=1}^n \delta_{Z_i}$$

==

z	z_1	z_2	$\dots$	z_n
$P(Z=z_i)$	$1/n$	$1/n$		$1/n$

$n \rightarrow \infty$



Lemma 2.3: Conditional exchangeability given a symmetric function

Let $Z_1, \dots, Z_n \in \mathcal{Z}$ be exchangeable, and let $f: \mathcal{Z}^n \rightarrow \mathcal{W}$ be a symmetric function, i.e., $f(z_1, \dots, z_n) = f(z_{\sigma(1)}, \dots, z_{\sigma(n)})$ for all $z_1, \dots, z_n \in \mathcal{Z}$ and all $\sigma \in \mathcal{S}_n$. Then $(Z_1, \dots, Z_n)$ is conditionally exchangeable given $f(Z_1, \dots, Z_n)$, in the sense that the conditional distribution

$$(Z_1, \dots, Z_n) \mid f(Z_1, \dots, Z_n)$$

is, almost surely, an exchangeable distribution.

$$\sum_{i=1}^n z_i = 2$$

Proof of Lemma 2.3

By definition of exchangeability, it suffices to verify that for any $\sigma \in \mathcal{S}_n$ and any measurable set A , the following statement holds almost surely:

$$\mathbb{P}((Z_1, \dots, Z_n) \in A \mid f(Z_1, \dots, Z_n)) = \mathbb{P}((Z_{\sigma(1)}, \dots, Z_{\sigma(n)}) \in A \mid f(Z_1, \dots, Z_n)).$$

Equivalently, we need to show that

$$\mathbb{P}((Z_1, \dots, Z_n) \in A, f(Z_1, \dots, Z_n) \in B) = \mathbb{P}((Z_{\sigma(1)}, \dots, Z_{\sigma(n)}) \in A, f(Z_1, \dots, Z_n) \in B)$$

for all measurable $A \subseteq \mathcal{Z}^n, B \subseteq \mathcal{W}$. This holds because

$$\begin{aligned} \mathbb{P}((Z_1, \dots, Z_n) \in A, f(Z_1, \dots, Z_n) \in B) &= \mathbb{P}((Z_{\sigma(1)}, \dots, Z_{\sigma(n)}) \in A, f(Z_{\sigma(1)}, \dots, Z_{\sigma(n)}) \in B) \\ &= \mathbb{P}((Z_{\sigma(1)}, \dots, Z_{\sigma(n)}) \in A, f(Z_1, \dots, Z_n) \in B), \end{aligned}$$

where the first step holds since $(Z_1, \dots, Z_n)$ is exchangeable, while the second step holds by symmetry of f .

Proof of Proposition 2.2

Since $\hat{P}_n$ is a symmetric function of the random variables $Z_1, \dots, Z_n$, by Lemma 2.3 below it holds almost surely that $Z_1, \dots, Z_n$ are exchangeable conditional on $\hat{P}_n$, and consequently, it holds almost surely that

$$\mathbb{P}(Z_i \in A \mid \hat{P}_n) = \mathbb{P}(Z_j \in A \mid \hat{P}_n),$$

for each $j \in [n]$ and every (measurable) $A \subseteq \mathcal{Z}$. Assuming this holds, we then have

$$\begin{aligned} \mathbb{P}(Z_i \in A \mid \hat{P}_n) &= \frac{1}{n} \sum_{j=1}^n \mathbb{P}(Z_j \in A \mid \hat{P}_n) = \mathbb{E} \left[\frac{1}{n} \sum_{j=1}^n \underbrace{\mathbb{1}\{Z_j \in A\}}_{\delta_{Z_j}(A)} \mid \hat{P}_n \right] \\ &= \mathbb{E} [\hat{P}_n(A) \mid \hat{P}_n] = \hat{P}_n(A), \end{aligned}$$

which proves the desired claim.

$\hat{P}_n$

$$\mathbb{P}(Z_j \in A \mid \hat{P}_n) = \mathbb{E}(\mathbb{1}\{Z_j \in A\} \mid \hat{P}_n)$$

Permutation tests

Let $\mathcal{P}$ be the set of all distributions on $\mathcal{Z}^n$, and let $\mathcal{P}_{\text{exch}} \subseteq \mathcal{P}$ be the subset of distributions for which exchangeability is satisfied. Consider a random vector $(Z_1, \dots, Z_n)$ drawn from some joint distribution P . We would like to perform a hypothesis test of

$$H_0 : P \in \mathcal{P}_{\text{exch}} \quad \text{versus} \quad H_1 : P \in \mathcal{P} \setminus \mathcal{P}_{\text{exch}}.$$

Lets say a large value of our test statistic $T(Z_1, \dots, Z_n)$ will indicate evidence against exchangeability.

$$p = \frac{\sum_{\sigma \in \mathcal{S}_n} \mathbb{1} \{T(Z_{\sigma(1)}, \dots, Z_{\sigma(n)}) \geq T(Z_1, \dots, Z_n)\}}{n!}$$

The following theorem shows it is a valid p-value

Theorem 2.4: Validity of the permutation test

For any function $T : \mathcal{Z}^n \rightarrow \mathbb{R}$, the p-value p defined in (2.3) satisfies $\mathbb{P}_P(p \leq \tau) \leq \tau$ for all $\tau \in [0, 1]$ and all $P \in \mathcal{P}_{\text{exch}}$.

it is common to avoid the computational burden of computing all $n!$ permutations by instead sampling a smaller number M of permutations $\sigma_1, \dots, \sigma_M \in \mathcal{S}_n$ uniformly at random,

$$p = \frac{1 + \sum_{m=1}^M \mathbb{1} \{T(Z_{\sigma_m(1)}, \dots, Z_{\sigma_m(n)}) \geq T(Z_1, \dots, Z_n)\}}{1 + M}. \quad (2.4)$$

Univariate regression

$$\{(x_i, y_i)\}_{i=1}^n \rightarrow \beta = 0$$

$$y_i = \alpha + \beta x_i + \varepsilon_i$$

$$\hat{\beta} = \frac{S_{yx}}{S_{xx}} = \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})}{\underbrace{\sum_{i=1}^n (x_i - \bar{x})^2}_{S_{xx}}}$$

$$T = \frac{\hat{\beta}}{se(\hat{\beta})} \quad se(\hat{\beta}) = \frac{s}{\sqrt{S_{xx}}}$$

$$s = \sqrt{\frac{SSE}{n-2}}$$

$$T \sim t_{n-2}$$

$$H_0: \beta = 0 \quad H_a: \beta \neq 0$$

Assume $H_0: \beta = 0$

$$y_i = \alpha + \varepsilon_i \quad x_i, \quad \varepsilon_i \perp x_i$$

X	Y	X	Y	
x_1	y_1	x_1	$y_{\sigma_1(1)}$	$\rightarrow T_1$
x_n	y_n	x_n	$y_{\sigma_1(n)}$	

X	Y
X_1	$Y_{\sigma(1)}^m$
$\vdots$	$\vdots$
X_n	$Y_{\sigma(n)}^m$

$\rightarrow T_m$

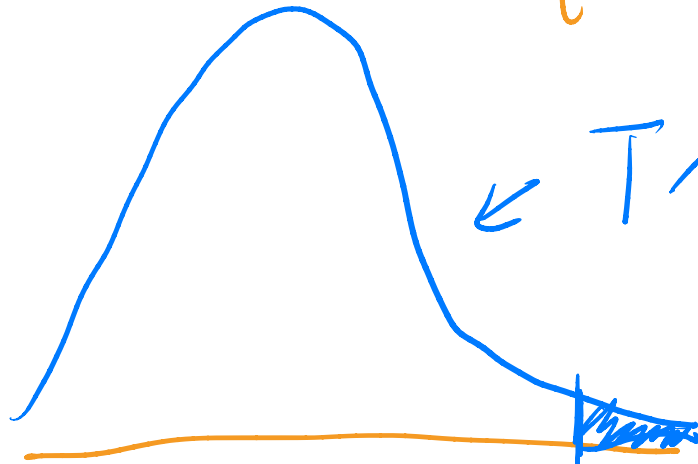
$$T_1 \text{ --- } T_m \rightarrow \hat{P}_m = \frac{1}{m} \sum_{i=1}^m \delta_{T_i}$$

$T \sim t_{n-2} \rightarrow$



$P(T \geq t) = \text{p-value}$

t



$T \sim \hat{P}_m$



t